

Cross-Talk Between One-Carbon Metabolism and Xenobiotic Metabolism: Implications on Oxidative DNA Damage and Susceptibility to Breast Cancer

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Published online: 27 July 2011
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Abstract The aim of this case–control study is to explore the role of aberrations in xenobiotic metabolism in inducing oxidative DNA damage and altering the susceptibility to breast cancer. Cytochrome P4501A1 (CYP1A1) m1 (OR: 1.41, 95% CI 1.08–1.84), CYP1A1 m4 (OR: 5.13, 95% CI 2.68–9.81), Catecholamine-*O*-methyl transferase (COMT) H108L (OR: 1.49, 95% CI 1.16–1.92), and glutathione *S*-transferase (GST) T1 null (OR: 1.68, 95% CI 1.09–2.59) variants showed association with breast cancer risk. Reduced folate carrier 1 (RFC1) 80A/CYP1A1 m1/CYP1A1 m4 and RFC1 80A/thymidylate synthase (TYMS) 5'-UTR 2R/methionine synthase (MTR) 2756G/COMT 108L genetic combinations were found to inflate breast cancer risk under the conditions of low dietary folate (345 ± 110 vs. 379 ± 139 $\mu\text{g/day}$) and low plasma folate (6.81 ± 1.25 vs. 7.09 ± 1.26 ng/ml) by increasing plasma 8-oxo-2'-deoxyguanosine (8-oxodG). This increase in 8-oxodG is attributed to low methionine (49.38 ± 23.74 vs. 53.90 ± 23.85 $\mu\text{mol/l}$); low glutathione (378 ± 242 vs.

501 ± 126 $\mu\text{mol/l}$) and GSTT1 null variant; and hypermethylation of CpG island of extracellular-superoxide dismutase (EC-SOD) (92.78 ± 11.49 vs. $80.45 \pm 9.86\%$), which impair *O*-methylation of catechol estrogens to methoxy estrogens, conjugation of glutathione to semiquinones/quinones and free radical scavenging respectively. Our results suggest cross-talk between one-carbon metabolism and xenobiotic metabolism influencing oxidative DNA damage and susceptibility to breast cancer.

Keywords Breast cancer · Extracellular-superoxide dismutase · Methylation · One-carbon metabolism · 8-oxo-2'-deoxyguanosine · Xenobiotic metabolism

Introduction

Breast cancer is rapidly emerging as the most common cancer among women both in developed and developing countries [1]. Since highly penetrant mutations account for small proportion of cases [2], several epidemiological studies were conducted that indicated the involvement of less penetrant genetic polymorphisms and carcinogens (both endogenous and exogenous) in the pathophysiology of breast cancer. Prolonged exposure to estrogen is widely accepted risk factor for breast cancer [3]. Estrogens can act as potential carcinogens either by stimulating the transcription of genes necessary for cell proliferation via the estrogen receptor [4] or by inducing oxidative DNA damage [5]. 17- β -estradiol is oxidized to 2- and 4-catechol estrogens in the presence of Phase I enzymes, i.e., CYP1A1 and CYP1B1 [6, 7]. Catechol estrogens on further oxidation form semiquinones and quinones, which are highly reactive species with a tendency to form adducts with DNA leading to mutagenic lesions such as 8-oxo-2'-deoxyguanosine

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(8-oxo-dG). Phase II enzymes, i.e., catecholamine methyltransferase (COMT) and glutathione *S*-transferase (GST) catalyze the detoxification of reactive oxygen species by *O*-methylation of catechol estrogens [8] and by conjugation of semiquinones and quinones with glutathione [9], respectively. Both Phase I and II enzymes being the integral part of xenobiotic metabolism help in biotransformation of lipophilic carcinogens to hydrophilic compounds to facilitate easy excretion. The availability of methyl group through *S*-adenosyl methionine (SAM) is crucial for COMT-mediated catalysis. SAM is the product of one-carbon metabolism.

Several studies have demonstrated that aberrations in xenobiotic and one-carbon metabolic pathways are documented risk factors for breast cancer [10–23]. Three polymorphisms in CYP1A1 were studied extensively in relation to breast cancer: m1, a T→C transition giving rise to MspI restriction site in the 3′-noncoding region [10]; m2, a substitution of valine for isoleucine at codon 462 in the heme-binding domain of exon 7 [11]; m4, a substitution of asparagine for threonine at codon 461 adjacent to the m2 polymorphism [12]. CYP1A1 m1 polymorphism was found to be associated with “high inducibility phenotype” resulting in higher levels of the allelic variants and elevated polycyclic aromatic hydrocarbon (PAH)-adducts [13]. This allelic variant was found to be a risk factor for breast cancer in Chinese [14] and no such association was observed in Caucasians and African-Americans [15]. CYP1A1 m2 variant, although has no direct influence on enzymatic activity, was found to be associated with more PAH-DNA adducts in lymphocytes [16]. The results of association studies conducted world-wide are conflicting in relation to m2 genotype with meta-analysis showing the lack of association [17]. Studies on CYP1A1 m3 and m4 polymorphisms are sparse [15]. G→A polymorphism in COMT causes valine to methionine substitution at codon 108 soluble (S)/158 (membrane-bound) COMT, generating alleles encoding high activity, thermostable (H) and low activity, thermolabile (L) forms of the enzymes [18]. L-form was found to be associated with higher levels of depurinating estrogen-DNA adducts that trigger mutagenesis and initiate cancer [19]. The association studies on COMT polymorphism were also contradictory [20, 21]. COMT catalysis is the connecting link between one-carbon and xenobiotic pathways. Deletions in GST T1 and GST M1 were also found to confer risk for breast cancer [22, 23].

Endogenous estrogens can become carcinogenic via formation of catechol estrogen quinones either due to increased CYP1A1 activity or due to decreased COMT/GST-mediated detoxification. Aberrations in one-carbon metabolism might influence the detoxification process of xenobiotic pathway by limiting the supply of methyl

moiety and glutathione for COMT-mediated conversion of catechol estrogens to methoxy estrogens and conjugation of quinones/semiquinones respectively. Recently we demonstrated that gene–gene interactions between RFC1 G80A/thymidylate synthase (TYMS) 5′-UTR 28 bp tandem repeat polymorphism/methylene tetrahydrofolate reductase (MTHFR) C677T polymorphisms of one-carbon metabolism determine the susceptibility to breast cancer, while cytosolic serine hydroxymethyltransferase (cSHMT) C1420T confers protection [24]. Further, we also observed increased oxidative stress in breast cancer subjects with low intake of dietary folate, low plasma folate and RFC1 G80A polymorphism and decreased oxidative stress in subjects with high dietary folate, high plasma folate, high glutathione, and cSHMT C1420T polymorphism [25]. In the current study, we explore the influence of one-carbon variants on xenobiotic metabolism and possible cross-talk between the variants of both the pathways in modulating oxidative DNA damage and breast cancer risk.

Materials and Methods

Subjects

For the study, we enrolled 595 subjects (342 consecutive breast cancer cases and 253 healthy controls) during the period of January 2009 to February 2011 at Nizam’s Institute of Medical Sciences, Hyderabad, India. The participation refusal rate was 0.88%. South Indian women who were newly diagnosed to have breast cancer based on mammography and histopathological investigations were enrolled as cases. The inclusion criteria were women in the age group of 18–70 years with confirmed breast cancer diagnosis. The clinicopathological characteristics of breast cancer cases were presented in Table 1. The exclusion criteria were women who had any other malignant or inflammatory disease and women who underwent surgical/radiation/chemotherapy. Healthy volunteers comprising of hospital staff and their relatives with no history of any benign or malignant disease were enrolled as controls. Breast health assessment of the control subjects was done based on breast imaging reporting and data system classification (based on mammography) to rule out the absence or likelihood of breast cancer and observed a score of 1, i.e., negative for breast cancer, on a scale of 1–5. None of the subjects were recruited in the absence of mammography findings and general health of each subject was evaluated by a physician. Personal interviews were conducted by trained personnel to collect information regarding age, body mass index, food frequency, tobacco, alcohol, active/passive smoking, use of oral contraceptives or hormone replacement therapy, age of menarche, age at first full term

Table 1 Clinicopathological characteristics of enrolled breast cancer cases

Clinicopathological characteristic	Number of subjects
Histological subtype	
Ductal	332
Lobular	5
Mixed (ductal and lobular)	4
Mucinous	1
Breast cancer grade (Scarff-Bloom-Richardson scale)	
Low grade (stage I)	68
Intermediate grade (stage IIa and IIb)	134
High grade (stage IIIa, IIIb, and IV)	140
Axillary lymph node involvement	
Yes	133
No	207
Distant metastasis	2
Immunohistochemical phenotypes	
Luminal A	110
Luminal B	55
Basal-like	60
LuminalA-HH	43
LuminalB-HH	31
HER2-enriched	43

pregnancy, parity, menopause status, and family history of breast cancer during recruitment process. The clinical parameters were obtained from patients' medical records with follow-up information provided by the Tumor Registry. The study protocol was approved by the Ethical committee of Nizam's Institute of Medical Sciences, Hyderabad, India. Informed consent was obtained from all the subjects. Venous blood samples were obtained from all the subjects in EDTA vacutainers prior to surgical or therapeutic intervention. Genomic DNA was isolated from all the samples using the standard DNA extraction protocol.

Dietary micronutrient assessment was done based on validated food frequency questionnaire. The details about the questionnaire, assessment, and validation were described elsewhere [24].

Estimation of Biochemical Parameters

Plasma 8-oxodG (Northwest Life Sciences specialties, USA) and estradiol (DRG International, Inc, Marburg, Germany) were measured using competitive ELISA kits as per the manufacturer's instructions. Ellman's method was used for the determination of glutathione [26]. Total plasma homocysteine was determined by using reverse phase HPLC method [27]. Plasma folate levels were estimated by using AxSYM folate kit (Abott Laboratories, USA).

Superoxide dismutase (SOD) activity was determined in RBC lysate by the method of McCord and Fridovich [28]. The superoxide ion was generated by xanthine–xanthine oxidase system. The assay mixture (final volume: 3 ml) contained 106 μ M EDTA, 16 μ M diethylene triamine penta acetic acid, 100 U catalase, 0.1 U xanthine oxidase, and 0.2 mM xanthine in 50 mM phosphate buffer, pH 7.8. To this assay mixture, 25 μ l of the RBC lysate was added and the reaction was initiated by the addition of 10- μ M cytochrome C. The reaction was monitored from 0 to 5 min at 550 nm. Activities were determined by calculating the percent inhibition of cytochrome C reduction.

Plasma methionine was estimated using reverse phase HPLC. In brief, 50 μ l of plasma sample was deproteinized with 50 μ l of 10% trichloro acetic acid. Protein-free supernatant was obtained by centrifugation at 13,000 rpm for 15 min. A 25- μ l aliquot of supernatant was subjected to vacuum drying. To the dried sample, 10 μ l of drying solution (methanol: 1 M sodium acetate: triethylamine in 2:2:1 ratio) was added and allowed to dry under vacuum. To the dried sample, 20 μ l of derivatizing solution (methanol:water:triethylamine:phenyl isothiocyanate in 7:1:1:1 ratio) was added and allowed to dry under vacuum. To the derivatized sample, 100 μ l of diluent (5 mM disodium hydrogen phosphate adjusted to pH, 7.4 with orthophosphoric acid and 5% acetonitrile) was added. Phenyl thiocarbamate derivatives of amino acids obtained by this procedure were subjected to reverse phase High performance liquid chromatography using Inertsil ODS-3 V (C/N 5020-01802, 5 μ m, 4.6 $\times$ 250 mm, S/N 8cI84300 from GL Sciences Inc, Japan) as stationary phase, binary gradient with reagent 1 (140 mM sodium acetate, 0.05% triethylamine adjusted to pH, 6.4 with glacial acetic acid, 6% acetonitrile in final volume) and reagent 2 (60% acetonitrile) as mobile phases on Waters HPLC system with λ_{max} at 254 nm. The resultant peaks were compared with a set of methionine standards.

Genetic Analysis of One-Carbon Metabolism

PCR-RFLP technique was used for the analysis of glutamate carboxypeptidase II (GCP2) C1561T, RFC1 G80A, cSHMT C1420T, TYMS 3'-UTR ins6/del6, MTHFR C677T, MTR A2756G, and methionine synthase reductase (MTRR) A66G polymorphisms as described earlier. PCR-AFLP technique was used for the analysis of TYMS 5'-UTR 28 bp tandem repeat polymorphism [24].

Genetic Analysis of Xenobiotic Metabolism

PCR-RFLP technique was used for the analysis of catecholamine methyltransferase (COMT) H108L, CYP1A1 m1, m2, and m4 polymorphisms. Multiplex-PCR was used for the analysis of deletions in GSTT1 and GSTM1 loci.

Extracellular-Superoxide Dismutase (EC-SOD) Promoter Methylation

Bisulfite Conversion of Genomic DNA

In an Eppendorf tube, 1 µg of genomic DNA was diluted to a final volume of 50 µl using TE buffer (tris 10 mM and EDTA 0.1 mM) and 5.7 µl of 3 M sodium hydroxide was added and incubated for 10 min at 37°C. Further, 33 µl of 16 mM hydroquinone and 530 µl of 4 M sodium bisulfite were added and incubated at 50°C overnight. The DNA was purified using nucleospin columns as per the supplier's instructions. The resultant mixture was treated with 5.7 µl of 3 M sodium hydroxide and incubated for 15–20 min at 37°C. The DNA was precipitated by adding a mixture of 17 µl of 7.5 M ammonium acetate and 450 µl of cold absolute alcohol and incubated at –80°C for 16–18 h. The supernatant was discarded after centrifugation (10,800 rpm, 4°C, 20 min) and pellet was washed with 70% alcohol and allowed to dry at ambient temperature. Dried pellet was dissolved in 35–45 µl of double distilled water.

Methylation-Specific PCR for EC-SOD

The bisulfite treated DNA was used as a template in two PCR sets, one using the primers specific for unmethylated promoter while the other using the primers specific for methylated promoter resulting in 284 and 279 bp products respectively. The corresponding sequences were located in 670–1428 bp of EC-SOD. A touch-down PCR strategy was adopted for annealing.

Statistical Analysis

Biochemical parameters were analyzed using student's *t* test (for two groups) or analysis of variance (ANOVA) (for more than two groups). Fisher's exact test was used for categorical variables to obtain crude odds ratios (ORs) and confidence intervals (CIs). Multiple logistic regression models were used to control the confounding effects and to obtain adjusted ORs and CIs. Spearman rank correlation was done to correlate biochemical parameters with genetic parameters. For all the statistical tests, www.statpages.org computational website was used. To explore gene–gene interactions across both the metabolic pathways, Multifactor Dimensionality Reduction (MDR Version 2.0 beta 8.2) analysis was performed by computing the genetic variables as 0, 1, and 2 based on number of variant alleles in each subject; 14 polymorphisms were used as predictor variables, i.e., X1, X2, ..., X14 and disease outcome was used as 'Class' for the analysis.

Results

All the polymorphisms in xenobiotic metabolic pathway were distributed in accordance with Hardy–Weinberg equilibrium. Among the six genetic polymorphisms tested, CYP1A1 m1, CYP1A1 m4, COMT H108L, and GSTT1 null were associated with 1.41-, 5.13-, 1.49-, and 1.68-folds increased risk for breast cancer after adjusting for confounding factors (Table 2).

Since one-carbon metabolism is likely to alter susceptibility to breast cancer by affecting biological reaction catalyzed by Phase II enzymes, the possibility of cross-talk between one-carbon and xenobiotic metabolic pathways was explored using multifactor dimensionality reduction analysis (MDR) that reduces the data dimension and to obtain the best possible genetic combinations that inflate breast cancer risk. This analysis showed significant gene–gene interactions between one-carbon and xenobiotic metabolic pathways, i.e., RFC 80A/CYP1A1 m1/CYP1A1 m4 (OR: 4.00, 95% CI 2.70–5.92) and RFC 80A/TYMS 5'-UTR 2R/MTR 2756G/COMT 108L (OR: 4.44, 95% CI 3.11–6.33) (Table 3).

Table 2 The distribution of six functional polymorphisms in breast cancer cases and controls

Polymorphism	0	1	2	MAF (%)	Adjusted OR (95% CI)
CYP1A1 m1					
Cases	168	125	49	32.60	1.41 (1.08–1.84)
Controls	140	98	15	25.29	
CYP1A1 m2					
Cases	221	108	13	19.59	0.71 (0.52–0.98)
Controls	144	99	10	23.52	
CYP1A1 m4					
Cases	274	68	0	9.94	5.13 (2.68–9.81)
Controls	240	13	0	2.57	
COMT H108L					
Cases	122	154	66	41.81	1.49 (1.16–1.92)
Controls	120	107	26	31.42	
GST T1 null					
Cases	255	87	–	12.72	1.68 (1.09–2.59)
Controls	205	48	–	9.48	
GST M1 null					
Cases	228	114	–	16.67	0.91 (0.62–1.33)
Controls	168	85	–	16.80	

0, 1, 2 number of minor alleles; MAF minor allele frequency; Adjusted OR odds ratio adjusted for age, body mass index, age at menarche, parity, menopause status, family history of breast cancer using logistic regression analysis; CI confidence interval; * Statistically significant

Note: GST T1 and GSTM1 '0' represents presence of allele while '1' represents null genotype

Table 3 Cross-talk between one-carbon and xenobiotic variants affecting breast cancer susceptibility

RFC-CYPm1-CYPm4	Cases/controls	Ratio	Class
0-0-1	14/2	7	1
0-0-0	44/57	0.77	0
0-1-1	11/2	5.5	1
0-1-0	30/34	0.88	0
0-2-0	13/4	3.25	1
1-0-1	12/2	6	1
1-0-0	60/70	0.86	0
1-1-1	54/49	1.10	0
1-2-1	8/2	4	1
1-2-0	19/9	2.11	1
2-0-1	7/0	α	1
2-0-0	31/9	3.44	1
2-1-1	15/8	1.88	1
2-2-1	1/0	α	1
2-2-0	8/0	α	1

Multifactor dimensionality reduction analysis (MDR) Odds ratio (95% CI): 4.00 (2.71–5.92); $\chi^2 = 51.6$, $P < 0.0001$; 0, 1, 2 represent number of minor alleles at the given loci; Class ‘1’ represents association with the disease while ‘0’ represent non-association

RFC 80A/TYMS 5'-UTR 2R/MTR 2756G/COMT 108L: MDR analysis based OR (95% CI) 4.44, 95% CI 3.11–6.33; $\chi^2 = 71.1$, $P < 0.0001$

As shown in Table 4, dietary folate, plasma folate, plasma methionine, plasma glutathione were lower and plasma homocysteine, estradiol and 8-oxo-2'-deoxyguanosine were higher in breast cancer cases compared to healthy controls.

Plasma 8-oxodG levels were found to be strongly influenced by xenobiotic variants as evidenced by consistent increase in 8-oxodG with increase in xenobiotic variant alleles in all combinations. One-carbon variants were found to augment the effect of xenobiotic variants in increasing 8-oxodG. Presence of three or more high-risk variant alleles in each metabolic pathway resulted in maximum increase in 8-oxodG levels (Table 5). Segregation of data based on dietary folate into subjects with folate below the median and subjects with folate above the median showed that folate

deficient group had elevated 8-oxodG (5.65 ± 1.82 vs. 4.36 ± 1.88 ng/ml, $P < 0.0001$).

Extracellular-superoxide dismutase (EC-SOD) was found to be hypermethylated (Fig. 1) in breast cancer cases compared to controls (92.78 ± 11.49 vs. $80.45 \pm 9.86\%$) thus resulting in lower SOD activity (Table 4). Dietary folate ($r = -0.25$) and cSHMT C1420T polymorphism ($r = -0.11$) showed inverse association with EC-SOD methylation while MTRR A66G polymorphism showed positive correlation ($r = 0.35$).

Discussion

The current study, to the best of our knowledge, is the first comprehensive analysis of one-carbon and xenobiotic variants in relation to oxidative DNA damage and breast cancer susceptibility. The data demonstrates independent association of CYP1A1 m1, CYP1A1 m4, COMT H108L, and GSTT1 null polymorphisms with breast cancer risk. In our recent study, we observed that RFC1 G80A, TYMS 5'-UTR tandem repeat and MTHFR C677T polymorphisms were independent risk factors in the same cohort [24]. In the current study, we demonstrate a cross-talk between one-carbon and xenobiotic metabolic pathways as evidenced MDR analysis showing significant gene–gene interactions, i.e., RFC 80A/CYP1A1 m1/CYP1A1 m4 and RFC 80A/TYMS 5'UTR 2R/MTR 2756G/COMT 108L in inflating breast cancer risk. One-carbon variants were found to augment the impact of xenobiotic variants in inducing oxidative DNA damage and one such mechanism for this synergistic impact is through hypermethylation of EC-SOD that inhibits SOD activity leading to impaired free radical scavenging.

Elevated estradiol levels coupled with CYP1A1 m1 and m4 polymorphisms might lead to increased catechol estrogen synthesis. Our results are consistent with Shin et al. [29] and Syamala et al. [30] in projecting CYP1A1 m1 as a risk factor for breast cancer. The findings of other studies were contradictory. Lack of association of CYP1A1 m2 polymorphism with breast cancer was consistent with

Table 4 Biochemical parameters in breast cancer cases and controls

Parameter	Cases	Controls	P value
Dietary folate ($\mu\text{g/day}$)	345 ± 110	379 ± 139	0.0009
Plasma folate (ng/ml)	6.81 ± 1.25	7.09 ± 1.26	0.007
Plasma homocysteine ($\mu\text{mol/l}$)	16.64 ± 7.82	12.45 ± 7.42	0.0001
Plasma methionine ($\mu\text{mol/l}$)	49.38 ± 23.74	53.90 ± 23.85	0.02
Plasma glutathione ($\mu\text{mol/l}$)	378 ± 242	501 ± 126	0.0001
Plasma estradiol (pg/ml)	129.8 ± 73.3	108.4 ± 57.6	0.0001
Plasma 8-oxodG (ng/ml)	5.68 ± 2.10	4.43 ± 1.85	0.0001
Superoxide dismutase activity (U/g Hb)	746 ± 370	1046 ± 380	0.0001

All the values are depicted in mean $\pm$ standard deviation format. All the P values in the given table are statistically significant

Table 5 Cross-talk between one-carbon and xenobiotic variants influencing plasma 8-oxodG (ng/ml) distribution

One-carbon variants	Xenobiotic variants				P_{trend}
	0	1	2	≥ 3	
0	3.2 ± 1.00 ($n = 9$)	3.60 ± 1.54 ($n = 26$)	5.05 ± 1.79 ($n = 23$)	5.91 ± 1.87 ($n = 19$)	0.0001
1	3.61 ± 0.96 ($n = 25$)	4.38 ± 1.89 ($n = 67$)	5.06 ± 2.08 ($n = 52$)	6.33 ± 2.11 ($n = 45$)	0.0001
2	4.13 ± 1.16 ($n = 32$)	4.77 ± 1.91 ($n = 62$)	5.57 ± 1.84 ($n = 58$)	6.74 ± 2.37 ($n = 45$)	0.0001
≥ 3	4.16 ± 1.15 ($n = 24$)	5.29 ± 1.89 ($n = 33$)	5.87 ± 1.71 ($n = 42$)	6.98 ± 2.24 ($n = 34$)	0.0001
P_{trend}	0.0004	0.006	0.18	0.03	

The data presented in the table is in mean $\pm$ SD. The data in *brackets* shows number of subjects, P_{trend} statistical significance of trend across number of gene variants. 0, 1, 2, 3 represent number of variant alleles at RFC1 80, TYMS 5'-UTR, MTHFR 677 loci (of one-carbon metabolism) and number of variant alleles at CYP1A1 m1, CYP1A1 m4, COMT 108, and GSTT1 null loci (of xenobiotic metabolism)

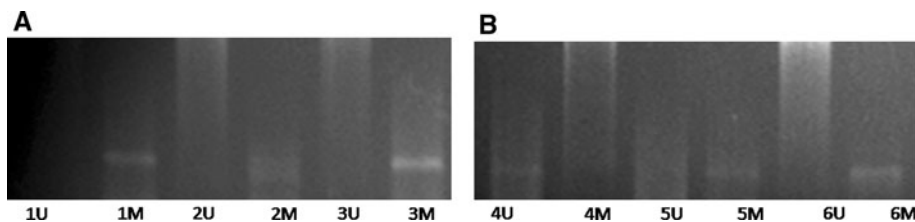


Fig. 1 Methylation specific PCR for extracellular-superoxide dismutase (EC-SOD): a representative gel picture of MS-PCR for EC-SOD showing prominent band with methylation-specific primers

(M) compared to primers specific for unmethylated promoter (U) in **a** breast cancer cases and **b** healthy controls

meta-analysis of Masson et al., case-control studies of Shin et al. [29] and Basham et al. [31]. There are limited studies on CYP1A1 m4 polymorphism as it was found to have very less frequency [32, 33]. COMT H108L polymorphism was more frequent in strongly ER-positive breast cancer than ER-negative breast cancer [34] while no association was observed in a study by Gaudet et al. [35].

Low dietary folate and plasma folate levels coupled with RFC1 G80A polymorphism would result in RBC folate deficiency thus decreasing the availability of functional folate essential for the synthesis of methionine and hence interfering with *O*-methylation of the catechol estrogens generated due to high inducible CYP1A1 (due to m1 and m4 variants). Presence of TYMS 5'-UTR 2R allele under conditions of folate deprivation might trigger DNA damage by inducing uracil misincorporation into the DNA due to reduced expression of TYMS. MTR A2756G in conjunction with RFC G80A, TYMS 5'-UTR 2R might interfere with methionine synthesis thus blocking the conversion of catechol estrogens to methoxyestrogens. The second level of detoxification is through conjugation of glutathione with semiquinones and quinones, which is also impaired as evidenced by low glutathione levels and also due to GSTT1 null variant. Redox cycling of quinone/semiquinone leads to the generation of reactive oxygen species (ROS), and adduct formation with DNA [36]. Since SOD activity is low, there is no effective scavenging of ROS thus resulting

in 8-oxodG lesions in DNA that induce mutagenicity (Scheme 1).

In the same cohort, we recently reported a strong association between the hyperhomocysteinemia and oxidative DNA damage [25]. Our earlier study shows dose-dependent association between homocysteine and DNA damage in vivo and in vitro [37]. Oikawa et al. [38] demonstrated similar association in human leukemia cell line HL-60, however, they observed no such increase in 8-oxodG with increase in homocysteine in hydrogen peroxide resistant clone HP-100 suggesting that the oxidative DNA damage is a result of superoxide generation. Mild increase in homocysteine (20 μ M) was found to induce piperidine-labile sites at the thymine residues and moderate to severe increase (100 μ M) was associated with 8-oxodG formation [38].

We observed lower SOD activity in breast cancer cases compared to controls. Since focal hypermethylation (of euchromatin) and global hypomethylation (of heterochromatin) are hallmarks of cancer, we hypothesized that aberrations of one-carbon metabolism might induce focal hypermethylation of SOD. Among the three distinct isoforms of SOD, Cu/Zn SOD is often constitutively expressed and not as easily inducible as other SODs thus serving as a “house keeper gene.” Epigenetic silencing of MnSOD in breast cancer cells and other transformed cells due to increased CpG methylation and decreased histone acetylation was reported earlier [39], however, no such changes in

The diagram illustrates the metabolic pathway of folate and its connection to estrogen metabolism and oxidative stress. The pathway begins with a decrease in Folyl polyglutamates, leading to Folyl monoglutamates. This is followed by a decrease in Tetrahydrofolate (THF), which is regulated by RFC G80A and SHMT. The resulting 5,10-Methylene THF is converted to 5-Methyl THF by MTHFR C677T. This conversion is coupled with the conversion of dUMP to dTMP, which is regulated by TYMS 5'-UTR 2R3R. The 5-Methyl THF is then converted to Methionine by MTR/MTRR. Methionine is converted to SAM, which is then converted to SAH. The conversion of SAM to SAH is coupled with the conversion of Homocysteine to Methionine. The diagram also shows the effects of the MTHFR C677T mutation on estrogen metabolism. The mutation leads to an increase in Estradiol, which is converted to Catechol estrogens by CYP1A1 m1 &m4. Catechol estrogens are then converted to Methoxy estrogens by COMT H108L. The diagram also shows that the mutation leads to a decrease in SAM, which is coupled with an increase in EC-SOD and hypermethylation. The diagram also shows that the mutation leads to an increase in ROS, which is coupled with an increase in 8-oxodG. The diagram also shows that the mutation leads to a decrease in GSH, which is coupled with a decrease in GSTT1 null and a decrease in GSH. The diagram also shows that the mutation leads to a decrease in Semiquinones/Quinones, which is coupled with a decrease in ROS and a decrease in 8-oxodG.

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graph TD
    A[↓ Folyl polyglutamates] --> B[Folyl monoglutamates]
    B --> C[↓ Tetrahydrofolate THF]
    C -- RFC G80A --> D[5,10-Methylene THF]
    C -- SHMT --> E[dUMP]
    E -- TYMS 5'-UTR 2R3R --> F[dTMP]
    D -- MTHFR C677T --> G[↓ 5-Methyl THF]
    G -- MTR/MTRR --> H[↓ Methionine]
    H --> I[↑ Homocysteine]
    I --> J[↑ SAM]
    J --> K[↓ SAM]
    K --> L[↑ EC-SOD hypermethylation]
    K --> M[↑ ROS]
    M --> N[↑ 8-oxodG]
    K --> O[↑ Catechol estrogens]
    O -- COMT H108L --> P[↓ Methoxy estrogens]
    O --> Q[Semiquinones/Quinones]
    Q --> R[↑ ROS]
    R --> N
    O --> S[↑ Estradiol]
    S -- CYP1A1 m1 &m4 --> O
    Q --> T[No conjugation]
    T -- GSTT1 null ↓ GSH --> Q
  
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The beneficial effects of folate and B₁₂ in restoring the SOD activity were demonstrated in a model of

Increased superoxide production as a result of hyperhomocysteinemia and during quinone/semiquinone formation, aberrant methylation of EC-SOD, low SOD activity and low total antioxidant status contribute to increased oxidative DNA damage. The excessive generation of ROS due to extensive granulocyte activation [43],

inflammatory cytokines [44], and malignant cells [45] as observed in advanced stages of cancer, might augment the oxidative stress further.

The major hallmarks of this study were: (i) the comprehensive analysis of one-carbon and xenobiotic variants with the help of MDR analysis that highlights the cross-talk between the variants of both the pathways thus enhancing our understanding of molecular pathophysiology of breast cancer and (ii) proposing an alternative mechanism, i.e., hypermethylation of EC-SOD to explain the impact of one-carbon variants in inducing oxidative DNA damage apart from well known phenomena, i.e., impaired conversion of catechol estrogens to methoxy estrogens and defective conjugation of glutathione to semiquinones/quinones due to limited supply of methyl moiety and glutathione. The limitation of this study was the assessment of dietary intake of B vitamins after the breast cancer diagnosis sensitive to recall bias.

To conclude, one-carbon variants influence xenobiotic metabolism by (i) limiting the supply of methyl moiety and glutathione and (ii) by inducing EC-SOD promoter hypermethylation thus increasing oxidative DNA damage and breast cancer risk. Further, the current study demonstrates the beneficial effects of high dietary folate in reducing oxidative DNA damage by restoring SOD activity and by enhancing free radical scavenging.

Acknowledgments This study was supported by the grant funded by Indian Council of Medical Research (ICMR), New Delhi (Ref No. 5/13/32/2007). Dr. V.K. Kutala is recipient of Ramanujan Fellowship awarded by Department of Science & Technology (DST), Government of India. We thank Dr. Annie Q Hasan, Head, Department of Molecular Medicine, Kamineni Hospitals, Hyderabad for critical revision of the manuscript in terms of genetic aspects.

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